AI SUPPLY CHAIN & PROCUREMENT — PROTOTYPE

VMI Automation

Autonomous tank monitoring, schedule parsing & order placement

WHY IT MATTERS

The supply chain team spends significant time every week managing one specific customer with a poor VMI profile: late Friday schedule emails, week-to-week volatility, frequent unplanned downtime, relatively small tanks, and shelf-life limits. This account has already absorbed real cost from returned trucks that wouldn't fit and aged-material returns — and real risk from multiple near run-outs.

WHAT IT DOES

- Ingests live tank telemetry every hour
- Parses schedule emails with rules-based parsing and LLM rescue on low-confidence cases
- Projects 7-day levels and auto-places EDI orders
- Fires live alerts before problems happen
- Logs every alert to build the feedback loop for ongoing tuning

LIVE ALERTS

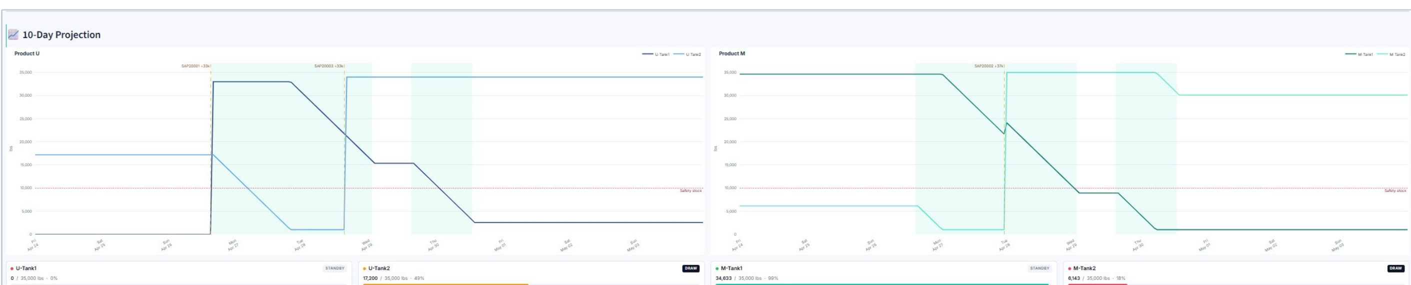
- Safety-stock breach projected
- Overfill on arriving truck
- Plant running off-schedule (3+ hrs)
- Lead-time shortfall
- Low-confidence schedule parse
- No schedule received by Fri 3 PM
- Late truck (3+ hrs overdue)

WORKFLOW

1. Hourly telemetry update triggers a fresh projection
2. Microsoft Graph checks the inbox for new schedules
3. Rules-based parser extracts run windows; LLM rescue handles low-confidence cases
4. Planner projects demand against dynamic targets
5. Load-entry PDF built and order placed via EDI
6. Every alert logged with context — feeding the feedback loop for ongoing tuning

DYNAMIC TARGET LEVELS

Reorder targets scale with projected weekly run hours. Light weeks (≤ 28 run hrs) target 15,000 lbs; heavy weeks (≥ 118 run hrs) target 27,000 lbs; intermediate weeks interpolate linearly. This reduces shelf-life exposure in slow weeks and run-out risk when the plant ramps up.



10-day projection from the live prototype — Product U and Product M tanks projected against safety-stock thresholds with scheduled EDI deliveries (shaded = inside lead-time window).

TECH & INTEGRATION STACK

Python Core platform	**Rules + LLM rescue** Schedule parsing	**Microsoft Graph** Email automation	**SAP + EDI** Verify & order

SAP is used to verify upcoming orders; orders are placed via EDI.

> **LIVE PROTOTYPE**
> **https://vmi-prototype.streamlit.app/**
> *Note: the prototype sleeps after inactivity. On first activation it may take a few minutes to wake up and load.*

PROTOTYPE SCOPE & PRODUCTION TARGETS

PROTOTYPE SCOPE — THIS VERSION

Simulation-driven dashboard with an advanceable clock, real IMAP email integration, JSON-backed state, and a regex-first schedule parser with optional LLM rescue. Projection, planner, and alerting logic are real and exercised against simulated tank levels and locally-assigned SAP order numbers.

PRODUCTION TARGETS — POST-PILOT

Microsoft Graph email, live tank telemetry, SAP order validation with full EDI acknowledgement loop, durable database with audit log, and role-based access control (RBAC).

IMPACT

Eliminate a repetitive manual task	**Reduce** runout & overfill risk	**Simplify** coverage within the team